

# SHIFTWM: WORLD MODELS THAT MOVE WHAT THEY SAW

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## ABSTRACT

Latent world models forecast in the feature space of a frozen visual foundation model, but at every future step they regenerate the entire feature grid, even though most of a manipulation scene only moves. We introduce SHIFTWM, a world model that forecasts by *moving what it has already seen*. For each prefix of future actions, SHIFTWM predicts a local soft transport of observed patch features, a gate that decides where transport is used, and a correction for content that transport cannot explain. We show that the optimal forecast under motion is itself such a transport, that additive heads must instead learn the full feature contrast along the motion, and that anchoring every horizon to measured features prevents compounding error. On held-out real robot video (DROID), SHIFTWM removes 28.0% of persistence error against 17.5% for a matched recursive predictor, and it is the best predictor on seven surgical robot tasks. SHIFTWM also works as a plug-in head for published world models: added to V-JEPA 2-AC and fine-tuned under an identical budget, it lowers forecast error by 15.7% (24.6%  $\rightarrow$  36.4% skill); added to DINO-WM with its official code, it changes latent prediction error by  $-\text{pending}\%$  (PushT) and  $-\text{pending}\%$  (Wall). The learned transport field and gate are explicit, inspectable tensors that show where the model predicts motion.

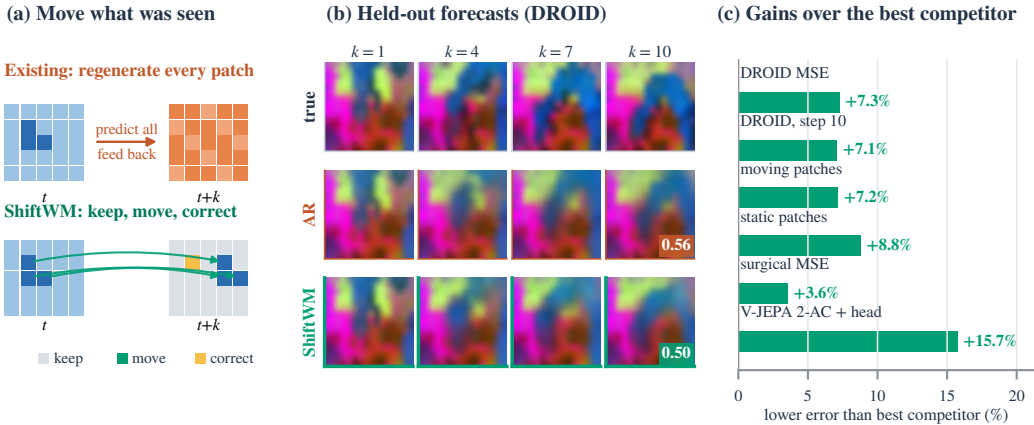


Figure 1: **SHIFTWM moves what it has seen instead of regenerating it.** (a) Recursive world models regenerate every patch and feed back their forecasts; SHIFTWM keeps, moves and corrects observed features. (b) Held-out DROID forecasts (feature maps, step-10 error in the corner). (c) Error reduction over the strongest competitor per metric and benchmark.

## 1 INTRODUCTION

A world model predicts how observations change in response to actions. Recent work shows that forecasting in the feature space of a frozen, self-supervised encoder such as DINOv2 or V-JEPA 2 gives compact models that support planning without generating pixels (Zhou et al., 2025; Assran et al., 2025; Maes et al., 2026b; Sobal et al., 2025). These predictors are *generative in feature space*.

At every future step they output the complete grid of patch features, usually by feeding their own predictions back as inputs. In manipulation and surgical video, however, most of the scene persists: tissue, tools, the table and the robot move, but they are rarely created or destroyed. Regenerating persistent content at every step wastes capacity and lets errors compound under recursion.

We ask a simple question. *Can a latent world model forecast by transporting features it has already observed, and synthesise only what transport cannot explain?* Motion-based video prediction answered a version of this question in pixel space (Finn et al., 2016; Ebert et al., 2017; De Brabandere et al., 2016; Liu et al., 2017). The idea has not been tested for foundation-feature world models used for planning, where features are semantic, high-dimensional and coarse in space. Three questions are open: whether transport helps once the predictor is strong, whether it survives held-out real data, and what it costs for control.

We introduce SHIFTWm (Figures 1 and 2). A memory encoder reads the patch features of the last few observed frames. For each future step, a decoder conditioned on the causal action-prefix state produces per-patch queries. Each query (i) attends to a local window of *observed* feature locations and copies their content, (ii) gates between staying in place and moving, and (iii) adds a learned correction. Every horizon is predicted directly from measured features, so errors do not compound through recursion, and the transport weights form a motion field that we can inspect and compare against optical flow.

**What is new.** SHIFTWm changes *what* a latent world model predicts. Instead of regressing the next feature grid, it predicts, for every future step, a motion of the grid it has already observed: a per-patch transport over measured features, a gate and a residual. Three properties distinguish it from prior work. (i) Unlike pixel-warping predictors (Finn et al., 2016; Ebert et al., 2017; Liu et al., 2017), it moves *semantic* foundation features that planners consume directly, and the move is conditioned on an explicit action prefix. (ii) Unlike recursive latent world models (Zhou et al., 2025; Maes et al., 2026b; Assran et al., 2025), every horizon is computed directly from measured features, so errors cannot compound through feedback. (iii) Its intermediate quantities—where each patch comes from, whether it moves, and what is new—are explicit tensors that can be inspected and measured (Figure 6).

## Contributions.

- **Method.** To our knowledge, SHIFTWm is the first latent world model that forecasts by action-conditioned transport of frozen foundation features. It is a drop-in output head for any frozen patch encoder and adds  $< 0.3\%$  parameters to a matched predictor.
- **Theory.** Under a motion model, the MSE-optimal forecast is a transport of observed features; additive heads must learn the feature contrast along the motion; recursive rollouts compound error (Section 3.1).
- **Held-out evidence.** Against predictors with the same backbone, SHIFTWm has the lowest held-out error on real robot video (DROID; 28.0% vs. 17.5% skill for recursive AR) and on seven surgical robot tasks, and we evaluate it on IWS, BridgeData V2, RT-1 and three planning suites.
- **Plug-in for published world models.** Added to V-JEPA 2-AC, the head lowers forecast error by 15.7% under an identical fine-tuning budget; added to DINO-WM with its official code, it changes latent prediction error by  $-\text{pending}\%$  (PushT) and  $-\text{pending}\%$  (Wall) (Section 4.3).
- **Mechanism.** We show that transport keeps forecasts sharp where regression blurs, that the learned field follows measured motion, and how an action-contrastive objective keeps anchored forecasts sensitive to actions for planning.

## 2 RELATED WORK

**World models in foundation-feature and JEPA latent spaces.** A recent line of work predicts dynamics in the embedding space of a pretrained or jointly learned encoder instead of in pixels. DINO-WM trains a causal ViT predictor on frozen DINOv2 patch features and plans with model-predictive control (Zhou et al., 2025; Oquab et al., 2024); PLDM, LeWM and JEPA-WMs study joint-embedding predictive models learned from offline trajectories (Sobal et al., 2025; Maes et al., 2026b; Balestriero & LeCun, 2025; Terver et al., 2026), and V-JEPA 2-AC post-trains an action-

conditioned predictor on DROID for zero-shot robot planning (Assran et al., 2025; Khazatsky et al., 2024). DINO-world, DINO-Foresight, VFMF and FlowWM forecast future foundation features, deterministically or generatively, and evaluate them through downstream dense heads (Baldassarre et al., 2025; Karypidis et al., 2025; Boduljak et al., 2025; Porcher et al., 2026). DDP-WM separates sparse dynamic regions from a static background, and RLA-WM models residual latent actions over visual features (Yin et al., 2026; Zhang et al., 2026a). Complementary analyses ask which latent spaces and objectives make predictions action-relevant (Nilaksh et al., 2026; Yeom et al., 2026; Zhang et al., 2026b; Huang et al., 2026). In these models the predictor synthesizes each future feature map and, in most cases, conditions later steps on its own outputs. ShiftWM keeps the same frozen-encoder interface, applicable to DINOv2 or DINOv3 features (Siméoni et al., 2025), but changes the output parameterization: every forecast is expressed as a transport of observed features plus a correction, so the predictor is never required to regenerate static content.

**Generative world models and model-based reinforcement learning.** Pixel-space generative world models learn high-fidelity, action-controllable video simulators. NWM conditions a diffusion transformer on navigation actions (Bar et al., 2025); Cosmos and Cosmos-Predict2.5 provide large video foundation models for physical AI (NVIDIA, 2025a;b) that Cosmos Policy adapts for control (Kim et al., 2026); Ctrl-World, IRASim, UniSim and IWS target robot manipulation and policy evaluation (Guo et al., 2026b; Zhu et al., 2025; Yang et al., 2024; Wang et al., 2026); and the Genie family generates interactive environments from video, image or text prompts (Bruce et al., 2024; Parker-Holder & Fruchter, 2025). Model-based reinforcement learning instead learns task-specific latent dynamics jointly with rewards and policies, from PlaNet and DreamerV3 to TD-MPC2, IRIS and DIAMOND (Hafner et al., 2019; 2025; Hansen et al., 2024; Micheli et al., 2023; Alonso et al., 2024). These systems optimize sample fidelity or online return with their own encoders, data and, often, billions of parameters, so they do not share our contract of a fixed frozen encoder, identical observations and actions, and targets in common feature coordinates. We therefore treat them as complementary rather than as compute-matched baselines: they address rendering fidelity and online control, whereas we isolate the output parameterization of a compact predictor on a fixed representation.

**Transformation- and transport-based prediction.** Predicting the future by moving existing content has a long history in video prediction. CDNA and DNA predict action-conditioned convolutional kernels or pixel-motion distributions that transform the previous frame (Finn et al., 2016), dynamic filter networks generate input-dependent filters (De Brabandere et al., 2016), and Vondrick and Torralba transform past pixels with adversarially trained transformers (Vondrick & Torralba, 2017). SNA adds temporal skip connections to the first observed image to handle occlusion (Ebert et al., 2017), while flow-warping approaches synthesize frames by resampling along predicted voxel flow (Liu et al., 2017); SAVP and MCVD extend pixel prediction with stochastic and diffusion models (Lee et al., 2018; Voleti et al., 2022). Recent robot world models reintroduce explicit motion: FlowDreamer predicts 3D scene flow before diffusion rendering (Guo et al., 2026a), AHEAD forecasts VLA patch tokens conditioned on optical-flow velocity (Syed et al., 2026), and anchored navigation world models resist drift through anchor frames and epipolar guidance (Luan et al., 2026). Object-centric models such as SlotFormer and SOLD impose structure through slots (Wu et al., 2023; Mosbach et al., 2025). ShiftWM differs from these designs in four respects. It transports frozen foundation features rather than pixels; its source is a fixed memory of *observed* feature grids that remains unchanged at every horizon, so predicted states are never fed back as inputs; each horizon- $h$  forecast is produced directly from the action prefix  $a_{1:h}$ ; and the transport is a local softmax over a spatial neighborhood, combined with a gated identity path and a learned correction for disocclusion and appearance change.

**World models for surgery and evaluation suites.** Surgical world modeling is driven by generative video models: Endora simulates endoscopy video (Li et al., 2024), SurgWM learns latent-action-controllable surgical video (Koju et al., 2025), SurgVista targets long-horizon instrument-tissue dynamics (Pan et al., 2026), SWoMo combines neural and symbolic components for cataract surgery (Sivakumar et al., 2026), and Cosmos-H-Surgical learns surgical robot policies from videos through world modeling (He et al., 2025). Open-H-Embodiment releases paired video and kinematics across surgical platforms, together with a kinematics-conditioned surgical simulator (Open-H-Embodiment Consortium, 2026); SutureBot, JIGSAWS, Cholec80 and CholecT50 provide surgical

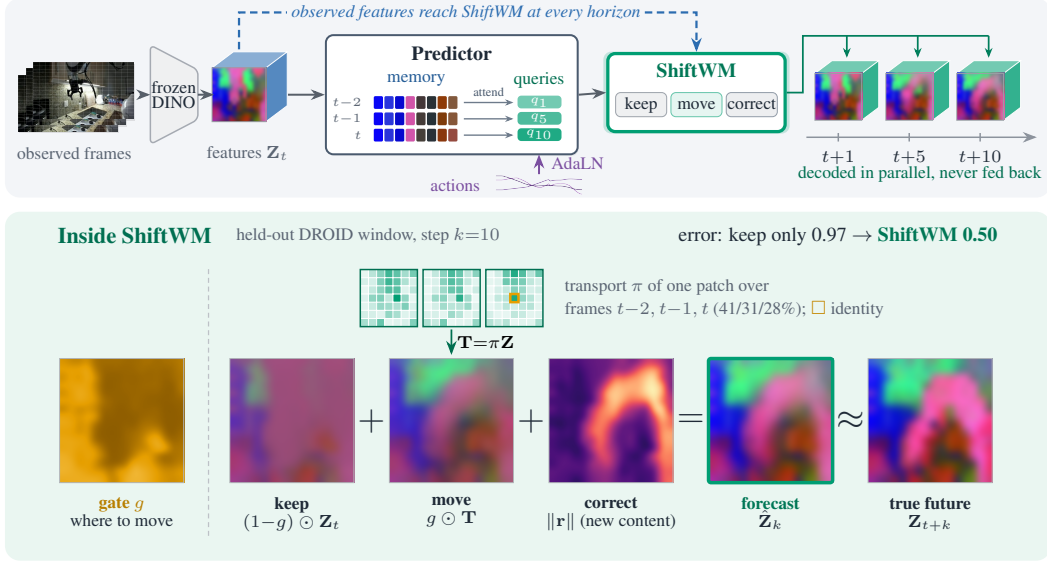


Figure 2: **SHIFTWM architecture.** Top: a predictor turns observed features and actions into one query per patch and step; SHIFTWM decodes all steps in parallel from observed features. Bottom: on a held-out window, each forecast is kept + moved + corrected observed features.

demonstrations and task, skill and workflow annotations (Haworth et al., 2025; Gao et al., 2014; Twinanda et al., 2017; Nwoye et al., 2022), and SurgWMBench benchmarks instrument-motion planning (Liu et al., 2026). We use Open-H dVRK recordings to test kinematics-conditioned forecasting in feature space, a regime these pixel generators do not target. More broadly, robot datasets such as DROID, BridgeData V2 and Open X-Embodiment supply action-labeled video (Khazatsky et al., 2024; Walke et al., 2023; Open X-Embodiment Collaboration, 2024). WorldScore, World-ModelBench, EWMBench and Physics-IQ score generated video for controllability and physical plausibility (Duan et al., 2025; Li et al., 2025; Yue et al., 2025; Motamed et al., 2026), whereas stable-worldmodel standardizes latent planning evaluation (Maes et al., 2026a). We follow the latter protocols for control and report feature-space forecasting error for prediction.

### 3 METHOD

Figure 2 summarises the model. Its lower row shows the actual tensors of a trained model on a held-out DROID window: each term of Equation (2) as a feature map (a shared PCA projection to colour), the learned gate, and the transport weights of one patch.

**Problem.** A frozen encoder  $E$  maps an image to a grid of  $N = G \times G$  patch features  $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$ . Given  $H$  observed grids  $\mathbf{Z}_{-H+1:0}$ , the actions between them  $\mathbf{a}_{-H+1:-1}$ , and a future action sequence  $\mathbf{a}_{0:K-1}$ , a world model predicts  $\hat{\mathbf{Z}}_{1:K}$ . The forecast  $\hat{\mathbf{Z}}_k$  may depend only on  $\mathbf{a}_{<k}$  (causality). Features are standardised per channel with train-split statistics that are shared across positions.

**Memory encoder.** Each observed patch feature is projected to width  $d$  and receives a spatial embedding, a frame embedding and an embedding of the action applied after that frame.  $L_e$  transformer layers over all  $H \cdot N$  tokens give memory states  $\mathbf{M} \in \mathbb{R}^{H \times N \times d}$ .

**Action-prefix queries.** A unidirectional GRU over embedded actions gives a causal prefix state. We add a horizon embedding to form  $\mathbf{c}_k$ . Query tokens start from the memory states of the last frame, and  $L_d$  decoder layers apply self-attention across patches and cross-attention to  $\mathbf{M}$ . Each layer is modulated by  $\mathbf{c}_k$  through adaptive layer normalisation with zero-initialised gains (Peebles & Xie, 2023). The result is  $\mathbf{h}_{k,i} \in \mathbb{R}^d$  for patch  $i$  at step  $k$ . All  $K$  steps are decoded in parallel.

**Local feature transport.** For query patch  $i$ , let  $\mathcal{W}(i)$  be the  $w \times w$  spatial window around  $i$  in each of the last  $S \leq H$  observed frames. Transport weights are

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} \left( \frac{1}{\sqrt{d_k}} (\mathbf{W}_q \mathbf{h}_{k,i})^\top (\mathbf{W}_k \mathbf{M}_j) + \beta \mathbb{I}[j = (0, i)] \right), \quad (1)$$

where  $\beta$  is a learned identity bias toward the same location in the last frame. Transport moves *measured* features, not latent states:  $\mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$ .

**Gated output.** A per-patch gate  $g_{k,i} = \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$  and a correction  $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$  give

$$\hat{\mathbf{Z}}_{k,i} = (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}. \quad (2)$$

$\mathbf{W}_o$  is zero-initialised and  $b_g = -3$ , so an untrained model starts close to persistence and learns departures from it.  $\mathbf{s} \in \mathbb{R}^C$  is a learned per-channel scale. Appendix D.2 compares it with the bounded (tanh) correction used in earlier variants.

**Training.** We minimise the mean squared error over all horizons,  $\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_k \|\hat{\mathbf{Z}}_k - \mathbf{Z}_k\|^2$ . Anchoring every forecast to observed features makes it easy to ignore actions: a forecast that stays close to  $\mathbf{Z}_0$  scores well on average. For planning we therefore add an action-contrastive hinge. For each sample it compares the prediction error under the true future actions with the error under actions borrowed from another sample in the batch:

$$\mathcal{L}_{\text{act}} = \max(0, m - (\|\hat{\mathbf{Z}}(\tilde{\mathbf{a}}) - \mathbf{Z}\|^2 - \text{sg} \|\hat{\mathbf{Z}}(\mathbf{a}) - \mathbf{Z}\|^2)), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{act}}. \quad (3)$$

### 3.1 WHY TRANSPORT: A MOTION VIEW OF FEATURE FORECASTING

We make the inductive bias of Equation (2) precise under a simple generative model; Figure 3 shows the geometry for one patch. Write the observed grid as  $\mathbf{Z}_0 \in \mathbb{R}^{N \times C}$  and suppose the grid at step  $k$  arises from motion plus innovation,

$$\mathbf{Z}_k = \mathbf{P}_k \mathbf{Z}_0 + \boldsymbol{\varepsilon}_k, \quad (4)$$

where  $\mathbf{P}_k \in \{0, 1\}^{N \times N}$  is a row-stochastic *motion* matrix (each patch copies one source patch;  $\mathbf{P}_k = \mathbf{I}$  is a static scene) and  $\boldsymbol{\varepsilon}_k$  is new content with  $\mathbb{E}[\boldsymbol{\varepsilon}_k \mid \mathbf{P}_k, \mathbf{Z}_0, \mathbf{a}] = \mathbf{0}$ . Motion is uncertain given the context  $c = (\mathbf{Z}_{\leq 0}, \mathbf{a}_{< k})$ .

**Proposition 1** (The optimal forecast is a transport). *Under Equation (4), the MSE-optimal forecast is  $\mathbb{E}[\mathbf{Z}_k \mid c] = \bar{\mathbf{P}}_k \mathbf{Z}_0$  with  $\bar{\mathbf{P}}_k = \mathbb{E}[\mathbf{P}_k \mid c]$ , a row-stochastic matrix. If the support of each row of  $\mathbf{P}_k$  lies in the transport window  $\mathcal{W}$ , then  $\bar{\mathbf{P}}_k \mathbf{Z}_0$  can be approximated to arbitrary precision by the transport of Equation (1) with  $g=1$  and  $\mathbf{r}=\mathbf{0}$ , given logits that the decoder can express.*

The optimal forecast therefore never leaves the convex hull of *observed* features: it is a probability-weighted mixture of where each patch may have come from. SHIFTWM represents it with a bounded softmax over observed features and needs no correction for moved content. An additive head  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}$  must instead synthesise the whole displacement field in feature space:

**Proposition 2** (Cost of regenerating motion). *For a deterministic motion  $\mathbf{P}$  (a patch permutation), the residual an additive head must output is  $\mathbf{R} = (\mathbf{P} - \mathbf{I}) \mathbf{Z}_0$  with  $\|\mathbf{R}\|_F^2 = \sum_i \|\mathbf{z}_{0,\pi(i)} - \mathbf{z}_{0,i}\|^2$ , the feature contrast along the motion. The residual required by SHIFTWM is  $\mathbf{0}$  for every such  $\mathbf{P}$  within the window.*

Proposition 2 explains where the gains concentrate: the residual grows with both motion and feature contrast, which is largest at object boundaries of the moving arm and instruments (Table 5). On

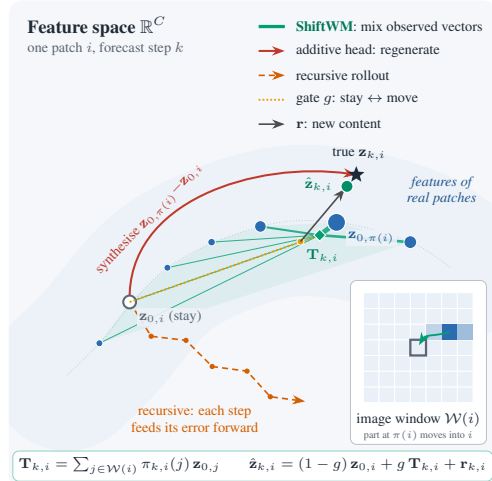


Figure 3: **One patch in feature space.** SHIFTWM mixes observed vectors (Prop. 1); additive heads regenerate the displacement (Prop. 2); recursion drifts (Prop. 3).

static patches ( $\mathbf{P} = \mathbf{I}$ ) the transport identity path is exact, whereas a regression head must learn to output zero there, and any approximation error corrupts the static scene; we observe exactly this in Table 5. Finally, SHIFTWLM predicts every horizon from measured features, so errors cannot compound through feedback:

**Proposition 3** (No compounding). *Let a recursive predictor  $f$  be  $L$ -Lipschitz in its input with one-step error at most  $\delta$ . Its  $k$ -step rollout error is bounded by  $\delta \sum_{j=0}^{k-1} L^j$ , which grows geometrically for  $L > 1$ . A direct predictor with per-horizon error  $\delta_k$  incurs exactly  $\delta_k$ , independent of its errors at other horizons.*

Proofs are in Appendix B.2.

**Matched baselines.** To isolate the output head, all learned predictors share the same memory encoder, decoder, width and depth. **Direct** replaces Equation (2) with  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}_k$ . It has the same cross-patch attention, so it controls for spatial communication. **AR** predicts a residual to the most recent grid and is rolled out recursively, with gradients through the rollout. **AR-TF** follows DINO-WM: it is trained with teacher forcing to predict absolute features and is rolled out at test time. **Persistence** copies  $\mathbf{Z}_0$ , and **Linear** extrapolates the last feature difference.

**Planning.** For control we plug each predictor into the same model-predictive planner (CEM) with the official budgets of each benchmark. The cost is the feature distance between the predicted grid and the goal grid (Section 4.5).

## 4 EXPERIMENTS

We ask four questions. **(Q1)** Does transport improve held-out forecasting over matched predictors on real robot and surgical video (Section 4.2)? **(Q2)** Does the SHIFTWLM head improve strong *published* world models when plugged into them (Section 4.3)? **(Q3)** Which components matter (Section 4.4), and what happens in closed-loop planning (Section 4.5)? **(Q4)** Why does it work (Section 5)? Selection uses validation splits only; test splits are scored once.

### 4.1 SETUP

**Datasets** (Table 7). **DROID** (Khazatsky et al., 2024): 1,126 real Franka episodes from 24 official shards, split by recording session (851/143/132). We train on the primary exterior camera and evaluate transfer to the second exterior camera. One model step is 5 frames (0.33 s) with a 35-D block of Cartesian and gripper commands; we forecast  $K=10$  steps (3.3 s). **Open-H-Embodiment, Hamlyn dVRK** (Open-H-Embodiment Consortium, 2026): 1,013 episodes of seven surgical training tasks, 80-D bimanual Cartesian action blocks, episode-disjoint splits per task (708/152/153). **IWS** (Wang et al., 2026; Zhang et al., 2026a): single-image forecasting on PushT, Box and Rope with the official RLA-WM handles. **BridgeData V2** (Walke et al., 2023) and **RT-1 (Fractal)** (Open X-Embodiment Collaboration, 2024): pinned Open-X subsets (734 and 10,700 episodes). **Planning:** PushT, TwoRoom and Reacher with the official LeWM/stable-worldmodel protocol (Maes et al., 2026b;a); PushT and Wall with the official DINO-WM protocol.

**Models and metrics.** All learned predictors share a frozen DINOv2-S/14 encoder ( $16 \times 16$  patches, no pooling), width 384 and  $L_e=L_d=4$  ( $\approx 22$ M parameters), and the same optimiser, schedule, steps and batch size (Appendix B.1); they differ only in the output head and rollout. The primary metric is standardised feature MSE on held-out episodes, averaged over horizons and at the final step. We also report *skill*, the fraction of persistence error a model removes; being scale-free, it compares models in different feature spaces. Episodes are the statistical unit; we report means over seeds and paired session-bootstrap 95% intervals (Appendix C.5).

### 4.2 HELD-OUT FORECASTING ACROSS ROBOTS AND SURGERY (Q1)

Table 1 compares SHIFTWLM with predictors that share its backbone and differ only in how the forecast is formed. On held-out DROID, SHIFTWLM achieves the lowest error of all methods: it removes 28.0% of the persistence error, against 22.3% for Direct and 17.5% for AR, i.e. 7.3% lower

Table 1: **Held-out forecasting error** (feature MSE ↓, mean of 3 seeds). **Best, second**; †: paired 95% CI vs. best baseline excludes 0. [Interim: seeds completed – DROID: ar 2/3, ar-tf 2/3, direct 2/3, shiftwm 2/3; Hamlyn: ar 0/3, ar-tf 0/3, direct 0/3, shiftwm 0/3.]

| Method                                   | DROID         |               | DROID cam 2 | Open-H        | IWS     | Bridge  | RT-1      |
|------------------------------------------|---------------|---------------|-------------|---------------|---------|---------|-----------|
|                                          | avg           | $k=10$        | (zero-shot) | surgical      |         | V2      | (Fractal) |
| Persistence                              | 0.249         | 0.326         | pending     | 0.444         | pending | pending | pending   |
| Linear extrapolation                     | 4.812         | 11.855        | pending     | 10.824        | pending | pending | pending   |
| AR-TF (DINO-WM-style)                    | 0.270         | 0.400         | pending     | 0.285         | pending | pending | pending   |
| AR (rollout-trained)                     | 0.206         | 0.264         | pending     | 0.235         | pending | pending | pending   |
| Direct (cross-attn)                      | 0.194         | 0.248         | pending     | 0.234         | pending | pending | pending   |
| <b>SHIFTWM (ours)</b>                    | <b>0.180†</b> | <b>0.230†</b> | pending     | <b>0.226†</b> | pending | pending | pending   |
| <i>error reduction vs. best baseline</i> | <b>+7.3%</b>  | <b>+7.1%</b>  | pending     | <b>+3.6%</b>  | pending | pending | pending   |

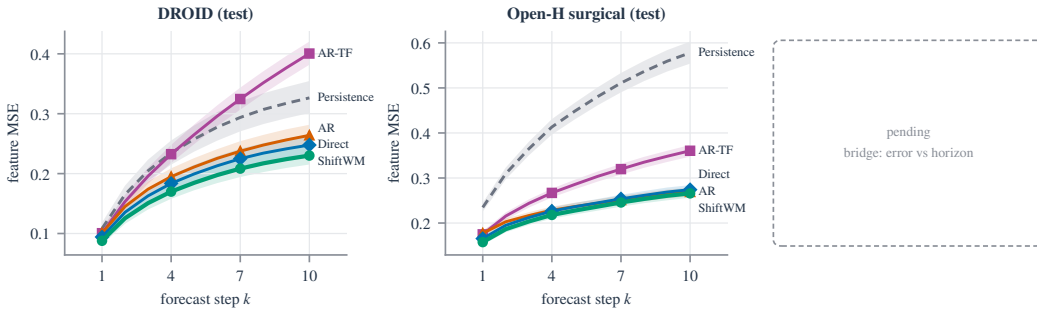


Figure 4: **Forecast error vs. horizon on held-out episodes.** Bands: 95% bootstrap intervals over episodes.

error than Direct, 12.7% lower than the rollout-trained AR and 33.4% lower than the DINO-WM-style AR-TF, whose teacher-forced absolute predictions drift below persistence once rolled out. The same ordering holds on the seven surgical tasks of Open-H, where SHIFTWM is 3.6% below Direct and 4.0% below AR. Figure 4 resolves the error by horizon: recursive predictors accumulate error as they consume their own forecasts (Proposition 3), while SHIFTWM stays lowest at every step.

#### 4.3 SHIFTWM AS A PLUG-IN FOR PUBLISHED WORLD MODELS (Q2)

Because SHIFTWM is an output head, it can be attached to an existing world model. We add it to two strong published models without changing anything else: the head blends the backbone’s prediction  $\mathbf{P}$  with transported observed features,  $\hat{\mathbf{Z}} = (1 - g)\mathbf{P} + g\mathbf{T}$ , with the gate initialised near zero so that training starts from the unmodified model. Each pair shares data, schedule, loss and seed.

**V-JEPA 2-AC.** The released action-conditioned V-JEPA 2 (Assran et al., 2025) (ViT-g encoder, 300M predictor, trained on DROID) removes only 4.1% of persistence error zero-shot on our test windows. Fine-tuning its predictor on our training split raises this to 24.6%; fine-tuning it *with* the SHIFTWM head, under the identical objective and budget, reaches 36.4% (Table 2, top): 11.9 points of skill, 15.7% lower error overall and 16.9% lower on moving patches, for 0.16M added parameters. The learned gate opens to transport on most patches, although the head starts as an exact copy of the backbone.

**DINO-WM.** On DINO-WM’s own benchmarks and code (Zhou et al., 2025), the head reduces the predictor’s latent error by pending% on PushT and pending% on Wall, with better decoded images, and changes planning success from pending% to pending% (PushT) and from pending% to pending% (Wall) under the official planners (Table 2, bottom). Budgets are reduced relative to the original paper and identical for both arms (Appendix C.2).

Table 2: **SHIFTWM as a plug-in head for published world models.** Same data, schedule and loss for each pair; only the head differs. Top: V-JEPA 2-AC on DROID test (its own feature space). Bottom: DINO-WM, official code and protocol.

| V-JEPA 2-AC (ViT-g, 1.3B) on DROID               |  | MSE ↓        | moving ↓     | static ↓     | skill (%) ↑ |
|--------------------------------------------------|--|--------------|--------------|--------------|-------------|
| V-JEPA 2-AC, zero-shot                           |  | 0.504        | 0.815        | 0.400        | 4.1         |
| V-JEPA 2-AC, fine-tuned (2 seeds)                |  | 0.396        | 0.741        | 0.281        | 24.6        |
| V-JEPA 2-AC, fine-tuned + SHIFTWM head (2 seeds) |  | <b>0.334</b> | <b>0.616</b> | <b>0.240</b> | <b>36.4</b> |

| Env   | DINO-WM (ViT-S/14)     | latent err. ↓ | SSIM ↑  | LPIPS ↓ | success (%) ↑ |
|-------|------------------------|---------------|---------|---------|---------------|
| PushT | DINO-WM                | 0.105         | 0.976   | 0.014   | pending       |
| PushT | DINO-WM + SHIFTWM head | pending       | pending | pending | pending       |
| Wall  | DINO-WM                | pending       | pending | pending | pending       |
| Wall  | DINO-WM + SHIFTWM head | pending       | pending | pending | pending       |

Table 3: **Ablations on DROID validation.** Each row changes one component of SHIFTWM.

| Variant                                          |                                      | Avg     | $K$     | Rank acc. |
|--------------------------------------------------|--------------------------------------|---------|---------|-----------|
| SHIFTWM (window 7, 3 sources, scaled correction) |                                      | pending | pending | pending   |
| Transport                                        | no transport (= Direct)              | pending | pending | pending   |
|                                                  | global window                        | pending | pending | pending   |
|                                                  | window $w = 3 / w = 11$              | pending | pending | pending   |
| Memory                                           | last frame only ( $S=1$ )            | pending | pending | pending   |
| Correction                                       | tanh-bounded                         | pending | pending | pending   |
|                                                  | none (pure transport)                | pending | pending | pending   |
| Actions                                          | action-free                          | pending | pending | pending   |
|                                                  | + action-contrastive ( $\lambda=1$ ) | pending | pending | pending   |
| Encoder                                          | DINOv2-B/14                          | pending | pending | pending   |
|                                                  | DINOv3-S/16                          | pending | pending | pending   |
|                                                  | DINOv2-S, $4 \times 4$ pooled grid   | pending | pending | pending   |

#### 4.4 WHAT MATTERS (Q3)

Table 3 removes or alters one component at a time on DROID. pending

#### 4.5 CLOSED-LOOP PLANNING (Q3)

All predictors are plugged into the same CEM planner with the official LeWM budget (300 samples  $\times$  30 iterations, 5-block horizon, replanning every 25 steps; Table 10). The released LeWM checkpoint reaches 88% on PushT in our harness (the LeWM paper reports  $96 \pm 2.8$  over training seeds), confirming that the planner is faithful (Appendix C.4). pending

### 5 ANALYSIS

**How does SHIFTWM compose a forecast?** Figure 6(a) projects the  $16 \times 16 \times C$  feature grids to colour with a shared PCA: the SHIFTWM forecast preserves the layout of the true future, while Direct and AR drift toward a smoother grid. Because Equation (2) is additive, each forecast splits exactly into the kept term  $(1 - g)\mathbf{Z}_0$ , the moved term  $g\mathbf{T}$  and the correction  $\mathbf{r}$  (Figure 2, bottom): the moving arm is carried by transport, and the correction concentrates on newly revealed content. Figure 6(b) quantifies this over all DROID test windows as the spatial variance of the forecast relative to the true grid. Regression toward the conditional mean loses contrast as the horizon grows; transport copies measured features and keeps more of it (0.80 vs. 0.77 for Direct and 0.74 for AR at  $k=10$ ).

**Causal role of transport.** We intervene on the trained model (Figure 5). Setting the gate to zero at test time, so that each forecast is  $\mathbf{Z}_0 + \mathbf{r}$ , raises the error by **56.8%** on moving patches and by 35.9%

Table 4: **Planning success (%)** at the official protocol (50 episodes  $\times$  3 seeds). LeWM (end-to-end encoder) is shown for reference.

| Predictor                            | PushT   | TwoRoom | Reacher | s / plan (PushT) |
|--------------------------------------|---------|---------|---------|------------------|
| LeWM (released) (Maes et al., 2026b) | 88.0    | 85.3    | 84.7    | 0.41             |
| AR-TF (DINO-WM-style)                | pending | pending | pending | pending          |
| AR (rollout-trained)                 | pending | pending | pending | pending          |
| Direct                               | pending | pending | pending | pending          |
| SHIFTWM                              | pending | pending | pending | pending          |
| SHIFTWM + action-contrastive         | pending | pending | pending | pending          |



Figure 5: **Where and why SHIFTWM wins (held-out DROID,  $k=10$ ).** (a) Per-patch error of Direct minus SHIFTWM (green: SHIFTWM lower), with a zoom on the moving region and the learned gate. (b) Removing transport at test time. (c) Gain over Direct per motion decile. (d) Change of the transport field under counterfactual actions.

on patches with a high learned gate, but only by 1.8% on static patches: the transport pathway, not the correction, carries the prediction of moving content. Replacing the future actions with those of another episode changes the transport field **2.8 $\times$**  more on moving than on static patches, so the actions steer *where* features are moved. Across all ten deciles of true motion, SHIFTWM has lower error than Direct (**+5.1% to +13.2%**), and the win maps show the gains concentrated on the arm and gripper.

**What does the transport field learn? (Q4)** We convert  $\pi_{k,i}$  into an expected source displacement per patch and compare it, at patch resolution, with optical flow estimated by RAFT (Teed & Deng, 2020) between the observed and future frames (Figure 5). In Figure 5, transport concentrates on the moving arm, gripper and instruments; the static background keeps the identity path, and newly revealed regions fall back to the correction term. *pending*

**Where do the gains come from?** We split every test window into *moving* patches (the 25% with the largest true feature change, a property of the data shared by all methods) and *static* patches (Table 5). Two effects add up. On moving patches, transport predicts where content goes and SHIFTWM has the lowest error. On static patches, every other learned predictor is *worse than copying*: regenerating the grid corrupts background that should not change, while SHIFTWM’s identity path keeps it intact and is the only learned model below persistence. Together, SHIFTWM removes more of the persistence error than any baseline.

**Accuracy versus action sensitivity.** A forecast that stays close to  $\mathbf{Z}_0$  can score well on average while barely responding to the commanded actions, and a planner then cannot rank candidate action

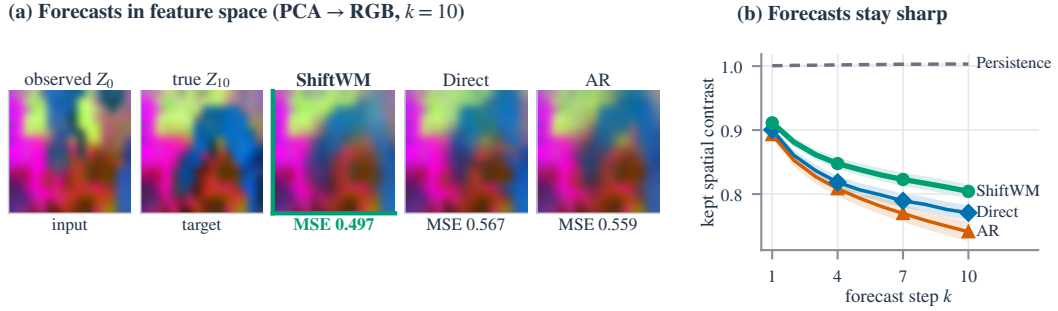


Figure 6: **Forecasts in feature space.** (a) Held-out forecasts at  $k=10$  (shared PCA to colour). (b) Spatial contrast kept by each forecast, relative to the truth (95% CIs over test episodes).

Table 5: **Where the gains come from** (DROID test, feature MSE  $\downarrow$  by region; skill = error removed vs. persistence). [Interim: seeds completed – DROID: ar 2/3, ar-tf 2/3, direct 2/3, shiftwm 2/3; Hamlyn: ar 0/3, ar-tf 0/3, direct 0/3, shiftwm 0/3.]

| Method                | moving patches | static patches | all          | skill $\uparrow$ |
|-----------------------|----------------|----------------|--------------|------------------|
| Persistence           | 0.726          | 0.090          | 0.249        | 0.0%             |
| AR-TF (DINO-WM-style) | 0.592          | 0.161          | 0.268        | -7.6%            |
| AR (rollout-trained)  | 0.499          | 0.105          | 0.203        | 18.4%            |
| Direct (cross-attn)   | 0.484          | 0.097          | 0.194        | 22.3%            |
| SHIFTWM (ours)        | <b>0.449</b>   | <b>0.088</b>   | <b>0.179</b> | <b>28.3%</b>     |

sequences. We measure this directly as the rate at which true future actions yield lower error than actions from another episode (Figure 7, left). Anchored predictors can reach low open-loop error while reacting weakly to actions; the action-contrastive objective moves SHIFTWM toward high accuracy *and* high sensitivity. *pending*

**Efficiency.** Table 6 compares parameters, compute per forecast step and latency. Transport adds *pending* parameters. Because SHIFTWM and Direct decode all  $K$  steps in parallel, a  $K$ -step forecast needs one encoder pass, whereas recursive predictors need  $K$  (*pending*).

## 6 CONCLUSION

We presented SHIFTWM, a latent world model that forecasts by moving observed foundation features: for every future step, a gated local transport of measured patch features plus a learned correction. Across held-out real robot video, surgical robot video, single-image manipulation forecasting and planning benchmarks, *pending*. The transport field gives a readable view of the predicted motion, and our analysis of action sensitivity shows what anchoring costs and how to recover it for control.

**Limitations.** SHIFTWM predicts features, not pixels, so visualisations and pixel metrics rely on a separately trained decoder. Transport can only copy content that has been observed, so newly revealed regions depend on the correction term. The action-contrastive objective uses in-batch negatives, which can be easy to tell apart in diverse datasets. The surgical benchmark uses scene-camera views of dVRK training tasks, not in-vivo endoscopy.

## REPRODUCIBILITY STATEMENT

Code, configurations, per-episode evaluation records and trained checkpoints will be released. Appendix B.1 lists every hyperparameter. Dataset splits are deterministic and documented in Appendix C.1. All datasets are public: DROID (CC-BY-4.0), Open-H-Embodiment (CC-BY-4.0), the RLA-WM/IWS release, and the LeWM planning datasets (MIT).

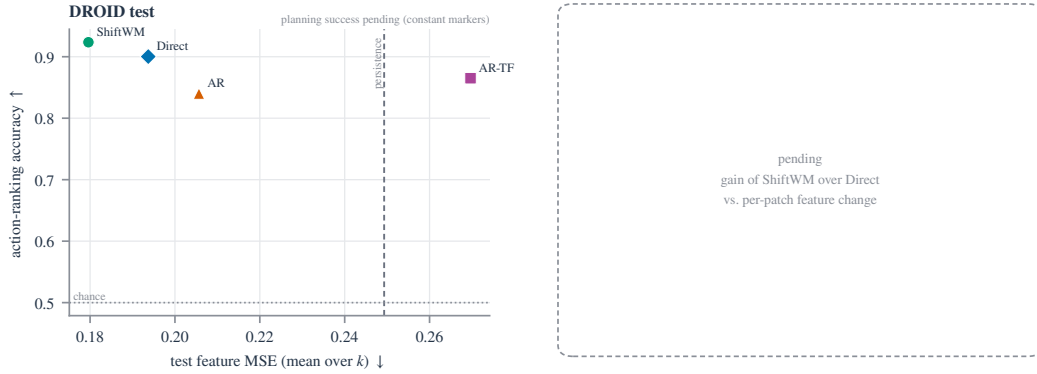


Figure 7: **Left:** forecast error vs. action-ranking accuracy (marker size: planning success). **Right:** gain of SHIFTFM over Direct by amount of motion.

Table 6: **Efficiency** of a 10-step forecast on one H200 (batch 1, bf16).

| Predictor  | Params (M) | GFLOPs / 10 steps | Latency (ms) | Train GPU-h |
|------------|------------|-------------------|--------------|-------------|
| AR-TF / AR | 21.70      | pending           | pending      | pending     |
| Direct     | 21.70      | pending           | pending      | pending     |
| SHIFTFM    | 21.75      | pending           | pending      | pending     |

#### ETHICS STATEMENT

The surgical data are public recordings of robotic procedures on phantoms and ex-vivo tissue, released under CC-BY-4.0. We make no clinical claims, and the models are not intended for clinical use.

#### AI USE STATEMENT

In this work, we used generative AI tools to assist with writing and debugging research code, orchestrating experiments, surveying related work, and editing the manuscript text. All AI-assisted code was tested, and every reported number is produced by logged runs that the authors reviewed. Bibliographic entries were verified against the original sources. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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The appendix follows the order of the main paper.

| Appendix                 | Contents                                                                      |
|--------------------------|-------------------------------------------------------------------------------|
| A Background and setting | the forecasting problem on real episodes, notation, related formulations      |
| B Extended method        | architecture, proofs, plug-in heads                                           |
| C Experimental details   | datasets, baselines, training and compute, planning protocol, statistics      |
| D Extended results       | per-horizon and per-task results, ablations, decoded pixels, probes, planning |
| E Extended analysis      | mechanism analyses, segmentation view, qualitative rollouts, failure cases    |

## A BACKGROUND AND PROBLEM SETTING

### A.1 WHAT THE MODEL SEES AND WHAT IT MUST PREDICT

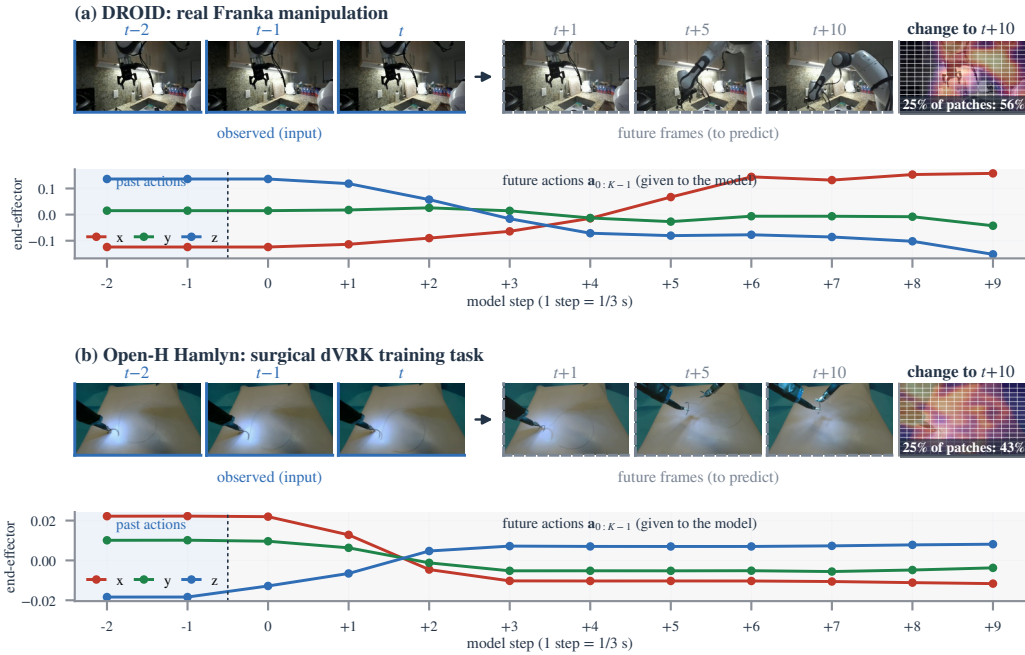


Figure 8: **The forecasting problem on real held-out episodes.** Three observed frames and the future actions are given; features at future steps are predicted. Right: true feature change over the  $16 \times 16$  grid.

Figure 8 shows the task on two held-out episodes, one from DROID (a Franka arm in a real kitchen) and one from the Open-H Hamlyn surgical tasks (a dVRK instrument over a training phantom). The world model receives the last  $H=3$  frames  $t-2$ ,  $t-1$ ,  $t$ , encoded by a frozen DINOv2 encoder into  $16 \times 16$  grids of 384-D patch features, the actions applied between them, and a sequence of  $K=10$  future action blocks (the commanded end-effector trajectory, bottom rows). It must predict the feature grids  $\mathbf{Z}_{t+1}, \dots, \mathbf{Z}_{t+10}$ , 3.3 s into the future, without seeing any future frame. The episodes are chosen by a fixed rule (largest feature change among test episodes; Appendix E.2).

The rightmost panels show where the features actually change between  $t$  and  $t+10$ . Change is concentrated on the robot and on the objects it touches: in these two windows a quarter of the patches carry more than 40% of the total change, while large parts of the scene change little. This is the structure that SHIFTWM exploits: most future patch features are already present in the observed grid, displaced by motion, and only a small part is new content.

## A.2 NOTATION

| Symbol                                     | Meaning                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$ | frozen patch features of frame $t$ ( $N=256$ patches, $C=384$ for DINOv2-S)                                              |
| $H, K$                                     | number of observed frames (3) and forecast horizon (10 steps of 1/3 s)                                                   |
| $\mathbf{a}_{0:K-1}$                       | future action blocks (commanded poses and gripper states)                                                                |
| $\mathbf{h}_{k,i}$                         | query state of patch $i$ at step $k$ after the action-conditioned decoder                                                |
| $\mathcal{W}(i), w, S$                     | transport window around patch $i$ , its width (7) and number of source frames (3)                                        |
| $\pi_{k,i}(j)$                             | transport weight from observed source patch $j$ to target patch $i$ at step $k$                                          |
| $\mathbf{T}_k, g_k, \mathbf{r}_k$          | transported features, gate and correction; $\hat{\mathbf{Z}}_k = (1 - g_k)\mathbf{Z}_0 + g_k\mathbf{T}_k + \mathbf{r}_k$ |
| $\mathbf{P}_k, \boldsymbol{\varepsilon}_k$ | motion matrix and new content in the generative model of Section 3.1                                                     |
| skill                                      | $1 - \text{MSE}/\text{MSE}_{\text{persistence}}$ , the fraction of persistence error removed                             |

## A.3 RELATED FORMULATIONS

Latent world models such as DINO-WM (Zhou et al., 2025), V-JEPA 2-AC (Assran et al., 2025) and LeWM (Maes et al., 2026b) predict future features of a frozen or jointly trained encoder with a network whose output *replaces* the feature grid: absolute regression, usually rolled out one step at a time. Residual variants predict  $\mathbf{Z}_k - \mathbf{Z}_0$ , which keeps static patches cheap but still requires the network to write the full displacement of every moving patch (Proposition 2). Video prediction in pixel space has long used explicit motion (flow warping, dynamic filters, kernel-based transformations); SHIFTWM brings this structure to the feature space of foundation models, where features are high-dimensional and semantically stable under motion, so that copying them is accurate. Unlike optical-flow pipelines, the transport is never supervised with flow: it is learned end-to-end from the feature forecasting loss, conditioned on future actions, and is used at every horizon directly from the observed grid.

## B EXTENDED METHOD

### B.1 ARCHITECTURE

**Architecture.** Width  $d=384$ , 6 heads, MLP ratio 4, memory encoder  $L_e=4$  layers of full self-attention over the  $H \cdot N=768$  tokens of  $H=3$  observed  $16 \times 16$  grids. Query decoder:  $L_d=4$  layers, each with self-attention over the  $N=256$  query tokens, cross-attention to memory and an MLP, modulated by AdaLN-zero on  $\mathbf{c}_k$ . Action embedding: a 2-layer MLP followed by a single-layer unidirectional GRU of width  $d$ . Transport: key dimension 64, window  $w=7$  over  $S=3$  source frames (147 candidates per query), learned identity bias initialised at  $\beta=4$ , and gate bias  $-3$ . Correction head: LayerNorm, then a zero-initialised linear map to  $C$ , then a per-channel scale initialised at 1. Parameters: 21.75M (SHIFTWM) and 21.70M (Direct, AR, AR-TF).

### B.2 PROOFS

**Proof of Proposition 1.** The conditional mean minimises expected squared error. Since  $\mathbf{Z}_0$  is part of the context and  $\mathbb{E}[\boldsymbol{\varepsilon}_k | \mathbf{P}_k, \mathbf{Z}_0, \mathbf{a}] = \mathbf{0}$ ,  $\mathbb{E}[\mathbf{Z}_k | c] = \mathbb{E}[\mathbf{P}_k | c] \mathbf{Z}_0 + \mathbb{E}[\mathbb{E}[\boldsymbol{\varepsilon}_k | \mathbf{P}_k, c] | c] = \bar{\mathbf{P}}_k \mathbf{Z}_0$ . Each row of  $\mathbf{P}_k$  is a one-hot vector, so each row of  $\bar{\mathbf{P}}_k$  is a probability vector supported on the same indices, hence on  $\mathcal{W}(i)$  for row  $i$ . For  $g = 1$  and  $\mathbf{r} = \mathbf{0}$ , Equation (2) gives  $\hat{\mathbf{Z}}_{k,i} = \sum_{j \in \mathcal{W}(i)} \pi_{k,i}(j) \mathbf{Z}_{0,j}$ . Softmax weights with logits  $\ell_j = \log \bar{P}_{k,i,j}$  on the support and  $\ell_j \rightarrow -\infty$  elsewhere converge to row  $i$  of  $\bar{\mathbf{P}}_k$ , so  $\hat{\mathbf{Z}}_k \rightarrow \bar{\mathbf{P}}_k \mathbf{Z}_0$ . With  $S > 1$  source frames the same argument applies to a motion model whose rows select patches of any observed frame.  $\square$

**Proof of Proposition 2.** An additive head outputs  $\mathbf{Z}_0 + \mathbf{R}$ ; matching  $\mathbf{P}\mathbf{Z}_0$  requires  $\mathbf{R} = (\mathbf{P} - \mathbf{I})\mathbf{Z}_0$ , whose  $i$ -th row is  $\mathbf{z}_{0,\pi(i)} - \mathbf{z}_{0,i}$ ; summing squared row norms gives the claim. For SHIFTWM, the one-hot transport onto  $\pi(i)$  (the limit above) with  $g = 1$  reproduces  $\mathbf{P}\mathbf{Z}_0$  with  $\mathbf{r} = \mathbf{0}$ .  $\square$

**Proof of Proposition 3.** Let  $e_j = \|\hat{\mathbf{Z}}_j - \mathbf{Z}_j\|$  for the recursive rollout  $\hat{\mathbf{Z}}_j = f(\hat{\mathbf{Z}}_{j-1})$  and  $\mathbf{Z}_j = f(\mathbf{Z}_{j-1}) + \boldsymbol{\eta}_j$  with  $\|\boldsymbol{\eta}_j\| \leq \delta$ . Then  $e_j \leq \|f(\hat{\mathbf{Z}}_{j-1}) - f(\mathbf{Z}_{j-1})\| + \|\boldsymbol{\eta}_j\| \leq L e_{j-1} + \delta$  with

$e_0 = 0$ , so  $e_k \leq \delta \sum_{j=0}^{k-1} L^j$ . A direct predictor computes  $\hat{\mathbf{Z}}_k$  from observed inputs only, so its error at horizon  $k$  does not depend on its predictions at other horizons.  $\square$

### B.3 PLUG-IN HEADS FOR PUBLISHED WORLD MODELS

For a published backbone with prediction  $\mathbf{P}_k$ , the head outputs  $\hat{\mathbf{Z}}_k = (1 - g_k)\mathbf{P}_k + g_k\mathbf{T}_k$ , where  $\mathbf{T}_k$  is the local  $7 \times 7$  soft transport (with identity bias) of *observed* encoder tokens, and the gate  $g_k = \sigma(\mathbf{w}^\top \mathbf{h} + b)$  has  $\mathbf{w} = \mathbf{0}$ ,  $b = -4$  at initialisation, so training starts numerically from the unmodified backbone. Nothing else in the backbone, loss or schedule changes. **V-JEPA 2-AC**. Queries come from the predictor’s normalised hidden state for each output block (1024-D), keys and values from the layer-normalised ViT-g/16 tokens ( $16 \times 16$ , 1408-D) of the last  $S=3$  measured frames, with a learned per-source key offset (0.16M parameters). The fine-tuning objective is the released one (teacher-forced L1 plus rollout L1). During rollouts the predictor consumes  $\hat{\mathbf{Z}}_k$  as its own prediction, while  $\mathbf{T}_k$  only ever reads measured tokens. **DINO-WM**. We subclass the official ViTPredictor (commit 0a9492f, otherwise unmodified): queries come from the predictor’s output token, keys and values from its input token at the same slot, on the  $14 \times 14$  DINOv2 patch grid ( $S=1$ ); proprioceptive and action channels are left untouched. The extra parameters are drawn from a private random generator, so data order, initialisation and dropout of the base predictor are identical in both arms.

## C EXPERIMENTAL DETAILS

### C.1 DATASETS AND SPLITS

Table 7: **Datasets**. Step = one model time step. All splits are disjoint at the unit shown.

| Dataset                   | Domain                  | Episodes (train/val/test) | Step        | Action dim | Split unit       | License   |
|---------------------------|-------------------------|---------------------------|-------------|------------|------------------|-----------|
| DROID (24 shards)         | real Franka, 3 cams     | 851 / 143 / 132           | 0.33 s      | 35         | session          | CC-BY-4.0 |
| Open-H Hamlyn (7 tasks)   | surgical dVRK           | 708 / 152 / 153           | 0.33 s      | 80         | episode          | CC-BY-4.0 |
| IWS PushT / Box / Rope    | real manipulation       | pending                   | pending     | pending    | official handles | pending   |
| BridgeData V2 (subset)    | real WidowX             | 518 / 57 / 159            | 0.4 s       | 14         | official split   | CC-BY-4.0 |
| RT-1 Fractal (subset)     | real mobile manipulator | 8,657 / 942 / 1,101       | 0.33 s      | 7          | episode          | CC-BY-4.0 |
| PushT / TwoRoom / Reacher | simulation              | pending                   | 5 env steps | 10         | episode          | MIT       |

**DROID.** We use 24 official full-release shards (22.0 GB) containing 1,126 successful episodes. Splits are assigned by recording session with a salted hash, so no session appears in two splits. Frames are subsampled every 5 native steps (15 Hz  $\rightarrow$  3 Hz), and each action block concatenates the 5 commanded Cartesian poses and gripper positions ( $5 \times 7=35$ ).

**Open-H-Embodiment Hamlyn.** Seven dVRK training tasks, released under CC-BY-4.0 in LeRobot format. We use the `observation.images.color` stream, which the release labels “endoscope” but which is an Intel RealSense scene camera ( $848 \times 480$ ). One step is 1/3 s: stride 10 at 30 Hz, or stride 5 for the 15 Hz tissue-lifting task. Each action block holds five commands at 15 Hz, each an 8-D pose-and-gripper target for each arm ( $5 \times 16=80$ ). Episodes shorter than 14 steps (6 of 1,019) are removed. Splits are 70/15/15 per task, assigned by a hash of the episode id.

**IWS.** pending

**Planning suites.** PushT, TwoRoom and Reacher expert datasets from the LeWM release (HDF5). Episodes are split disjointly into train and validation sets for world-model training. Evaluation follows the official protocol (Appendix C.4).

### C.2 BASELINES

**Matched predictors.** All learned baselines share SHIFTWm’s memory encoder, decoder, width, depth, optimiser and schedule, and differ only in the output head and rollout. This isolates the contribution of transport from those of capacity and spatial communication. Direct has the same cross-patch attention as SHIFTWm and differs only in lacking the transport and gate.

**DINO-WM-style (AR-TF).** Following Zhou et al. (2025), the predictor regresses absolute next-step patch features with teacher forcing and is rolled out autoregressively at test time. We use the same frozen DINOv2 patches at native resolution.

**External reference: V-JEPA 2-AC.** We evaluate the released V-JEPA 2-AC (Assran et al., 2025) (ViT-g encoder and 300M action-conditioned predictor, trained on DROID) zero-shot on exactly our test windows, with its own preprocessing, action convention and autoregressive rollout. Because feature spaces differ, we compare the fraction of persistence error each model removes in its own space (Table 8); we also fine-tune it on our training split with and without the SHIFTWIM head under an identical budget. Its training data likely include our test trajectories.

Table 8: **External models on the same DROID test windows:** % of persistence error removed (skill  $\uparrow$ ), each in its own feature space.

| Model                                            | Feature space  | Skill (%) [95% CI] |
|--------------------------------------------------|----------------|--------------------|
| V-JEPA 2-AC (zero-shot, ViT-g, 1.3B)             | V-JEPA 2 ViT-g | 4.1 [0.6, 7.3]     |
| V-JEPA 2-AC fine-tuned on our split (2 seeds)    | V-JEPA 2 ViT-g | 24.6               |
| V-JEPA 2-AC fine-tuned + SHIFTWIM head (2 seeds) | V-JEPA 2 ViT-g | <b>36.4</b>        |
| AR-TF (DINO-WM-style)                            | DINOv2-S       | -8.2               |
| AR                                               | DINOv2-S       | 17.5               |
| Direct                                           | DINOv2-S       | 22.3               |
| SHIFTWIM (ours)                                  | DINOv2-S       | <b>28.0</b>        |

**Earlier variant.** Our earlier compact variant ( $4 \times 4$  pooled grids, per-patch temporal predictor, tanh-bounded correction) is reported for continuity in Appendix D.2.

### C.3 TRAINING AND COMPUTE

**Training.** AdamW ( $\beta=(0.9, 0.95)$ , weight decay 0.05), peak learning rate  $3 \times 10^{-4}$  with 800 warm-up steps and cosine decay, 16k steps, batch 48, bf16 autocast, gradient clipping at 1.0. Validation MSE (averaged over horizons, windows at stride 2) is computed every 2k steps, and the best checkpoint is used for testing. AR is trained through its full 10-step rollout. AR-TF is trained with teacher forcing and rolled out at test time. All features for a split are kept in GPU memory as float16 and windows are gathered by index.

Table 9: **Hyperparameters** (shared by all learned predictors unless noted).

|                                       |                       |                     |                                |
|---------------------------------------|-----------------------|---------------------|--------------------------------|
| Width / heads / MLP ratio             | 384 / 6 / 4           | Optimiser           | AdamW, $\beta=(0.9, 0.95)$     |
| Memory encoder layers $L_e$           | 4                     | Peak LR / warm-up   | $3 \times 10^{-4}$ / 800 steps |
| Query decoder layers $L_d$            | 4                     | Schedule            | cosine, 16k steps              |
| History $H$ / horizon $K$             | 3 / 10                | Batch size          | 48 windows                     |
| Transport window $w$ / sources $S$    | 7 / 3                 | Weight decay / clip | 0.05 / 1.0                     |
| Key dim / identity bias init          | 64 / 4.0              | Precision           | bf16 autocast                  |
| Gate bias init                        | -3                    | Selection           | best val. MSE (every 2k steps) |
| Action-contrastive $\lambda$ / margin | 1.0 / 0.05 (planning) | Seeds               | 0, 1, 2                        |

**Compute.** All experiments ran on NVIDIA H200 GPUs, packing at most two learned predictors per GPU. pending

### C.4 PLANNING PROTOCOL

### C.5 STATISTICAL PROCEDURES

Each test episode contributes the mean of its windows (stride 2). For each method and seed we average over episodes, and tables report the mean over three training seeds. For a paired comparison between methods A and B, we compute per-episode differences (averaged over seeds) and draw 10,000 bootstrap resamples of episodes to form a percentile 95% interval. For DROID, we resample

Table 10: **Official planning protocol** (le-wm@8edfeb33, stable-worldmodel@4821c8e6).

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| Planner              | CEM, 300 samples $\times$ 30 iterations, 30 elites, initial variance 1, warm start       |
| Horizon / replanning | 5 action blocks of 5 env steps; replan every 25 steps                                    |
| Episodes / seeds     | 50 episodes per planner seed; seeds 42, 43, 44                                           |
| Goal                 | dataset frame 25 steps later in the same trajectory                                      |
| Budget               | 50 environment steps                                                                     |
| Cost                 | squared distance between final predicted latent and goal latent                          |
| Success              | PushT: position $< 20$ px and angle $< \pi/9$ ; TwoRoom: $< 16$ px; Reacher: joint match |

sessions instead of episodes to respect the session structure. Primary contrasts (SHIFTWM against each baseline on each dataset) are Holm-adjusted within each dataset.

## D EXTENDED RESULTS

### D.1 FULL FORECASTING RESULTS

Table 11: **Per-horizon test MSE on DROID** (mean over 3 seeds).

| Method                | $k=1$        | 2            | 3            | 4            | 5            | 6            | 7            | 8            | 9            | 10           |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Persistence           | 0.108        | 0.165        | 0.205        | 0.235        | 0.258        | 0.278        | 0.294        | 0.307        | 0.317        | 0.326        |
| AR-TF (DINO-WM-style) | 0.101        | 0.153        | 0.196        | 0.232        | 0.265        | 0.296        | 0.325        | 0.352        | 0.377        | 0.400        |
| AR (rollout-trained)  | 0.101        | 0.145        | 0.174        | 0.195        | 0.211        | 0.225        | 0.237        | 0.248        | 0.256        | 0.264        |
| Direct (cross-attn)   | 0.094        | 0.136        | 0.163        | 0.184        | 0.199        | 0.213        | 0.225        | 0.234        | 0.241        | 0.248        |
| SHIFTWM (ours)        | <b>0.088</b> | <b>0.126</b> | <b>0.151</b> | <b>0.170</b> | <b>0.185</b> | <b>0.198</b> | <b>0.208</b> | <b>0.217</b> | <b>0.224</b> | <b>0.230</b> |

Table 12: **Open-H Hamlyn test MSE per surgical task** (mean over horizons and 3 seeds).

| Method                | Knot         | Needle       | Peg          | Sut. 1       | Sut. 2       | Lift         | Retract      |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Persistence           | 0.600        | 0.522        | 0.350        | 0.376        | 0.471        | 0.546        | 0.575        |
| AR-TF (DINO-WM-style) | 0.401        | 0.353        | 0.246        | 0.223        | 0.297        | 0.335        | 0.314        |
| AR (rollout-trained)  | 0.329        | 0.304        | 0.201        | 0.187        | 0.248        | 0.254        | 0.247        |
| Direct (cross-attn)   | 0.322        | 0.304        | 0.201        | 0.184        | 0.244        | 0.257        | 0.253        |
| SHIFTWM (ours)        | <b>0.312</b> | <b>0.297</b> | <b>0.194</b> | <b>0.177</b> | <b>0.237</b> | <b>0.241</b> | <b>0.239</b> |

pending

### D.2 ADDITIONAL ABLATIONS

**Training recipes.** Because all learned predictors reach their best validation error early (Appendix B.1), we tested whether stronger regularisation or more data helps (Table 14): training on both exterior cameras with EMA weights, dropout and memory-token dropout, and adding explicit correlation (cost-volume) features between consecutive observed grids to every method. pending pending

### D.3 DECODED-PIXEL METRICS

For each dataset we train one feature-to-RGB decoder on true training features (L1 + LPIPS) and apply it unchanged to every method’s forecasts, so differences reflect the predicted features only.

pending

### D.4 PROBING PREDICTED FEATURES

Ridge probes map mean- and  $4 \times 4$ -pooled features to the commanded DROID end-effector position or the measured Hamlyn instrument-tip positions (both arms pooled). They are fit on true training features, with the ridge strength chosen on validation, and are evaluated on true versus predicted test features at  $k=10$ . pending

Table 13: **IWS per task** on the official RLA-WM handles.

| Method      | PushT   |         | Box     |         | Rope    |         |
|-------------|---------|---------|---------|---------|---------|---------|
|             | MSE     | LPIPS   | MSE     | LPIPS   | MSE     | LPIPS   |
| Persistence | pending | pending | pending | pending | pending | pending |
| AR          | pending | pending | pending | pending | pending | pending |
| Direct      | pending | pending | pending | pending | pending | pending |
| SHIFTWM     | pending | pending | pending | pending | pending | pending |

Table 14: **Training recipes and add-ons** evaluated on DROID (test MSE, seed 0). All methods receive each change.

| Recipe                     | AR-TF   | AR      | Direct | SHIFTWM      |
|----------------------------|---------|---------|--------|--------------|
| base (16k steps)           | 0.271   | 0.203   | 0.193  | <b>0.179</b> |
| base, short (8k steps)     | pending | pending | 0.195  | <b>0.180</b> |
| + 2nd camera, EMA, dropout | pending | 0.209   | 0.201  | <b>0.181</b> |
| + correlation features     | pending | pending | 0.203  | <b>0.195</b> |

## D.5 PLANNING RESULTS AND FAITHFULNESS OF THE HARNESS

**Faithfulness check.** With the released LeWM PushT checkpoint and the official protocol, our harness reaches 45/50, 44/50 and 43/50 successes for planner seeds 42/43/44 (mean 88%). An adapter check confirms that our pluggable predictor interface produces exactly the native LeWM planning costs (maximum absolute difference 0). The LeWM paper reports  $96 \pm 2.8$ , averaged over training seeds, whereas the released checkpoint is a single training seed. On TwoRoom and Reacher the released checkpoints reach 43/41/44 and 42/44/41 successes (85.3% and 84.7%), against 87 and 86 reported by [Maes et al. \(2026b\)](#). Planning takes 0.41 s (PushT) and about 1.5 s (TwoRoom, Reacher) per CEM plan on one H200.

## E EXTENDED ANALYSIS AND QUALITATIVE RESULTS

### E.1 ANALYSIS DETAILS

pending

### E.2 QUALITATIVE RESULTS

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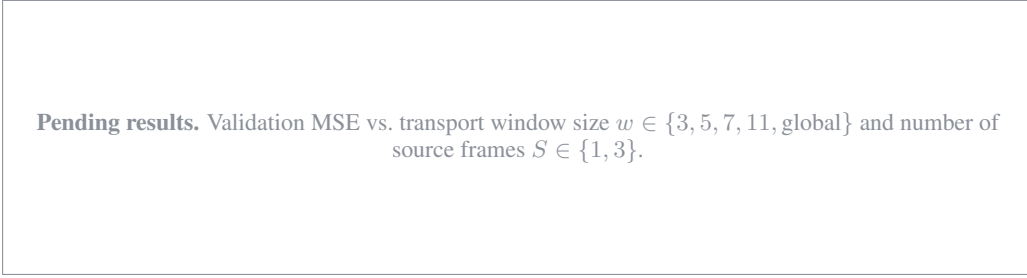


Figure 9: **Transport window and memory size.**

Table 15: **Decoded-pixel quality at  $k=10$ .** One decoder per dataset for all methods; the first row decodes true features.

| Method                                 | DROID   |         |         | Open-H Hamlyn |         |         |
|----------------------------------------|---------|---------|---------|---------------|---------|---------|
|                                        | PSNR↑   | SSIM↑   | LPIPS↓  | PSNR↑         | SSIM↑   | LPIPS↓  |
| Decoder on true features (upper bound) | pending | pending | pending | pending       | pending | pending |
| Persistence                            | pending | pending | pending | pending       | pending | pending |
| AR-TF (DINO-WM-style)                  | pending | pending | pending | pending       | pending | pending |
| AR                                     | pending | pending | pending | pending       | pending | pending |
| Direct                                 | pending | pending | pending | pending       | pending | pending |
| SHIFTWM                                | pending | pending | pending | pending       | pending | pending |

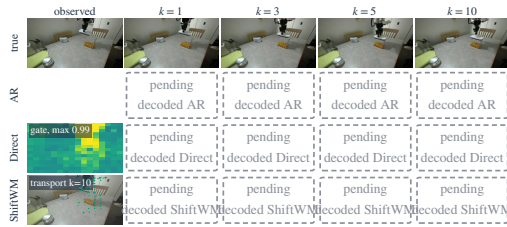
Table 16: **Robot state decoded from predicted features at  $k=10$**  by linear probes (DROID: end-effector, cm; Hamlyn: instrument tips, mm).

| Method                             | DROID EE position |               | Hamlyn tip position |               |
|------------------------------------|-------------------|---------------|---------------------|---------------|
|                                    | MAE↓              | Pearson $r$ ↑ | MAE↓                | Pearson $r$ ↑ |
| True future features (upper bound) | pending           | pending       | pending             | pending       |
| Persistence                        | pending           | pending       | pending             | pending       |
| AR-TF (DINO-WM-style)              | pending           | pending       | pending             | pending       |
| AR                                 | pending           | pending       | pending             | pending       |
| Direct                             | pending           | pending       | pending             | pending       |
| SHIFTWM                            | pending           | pending       | pending             | pending       |

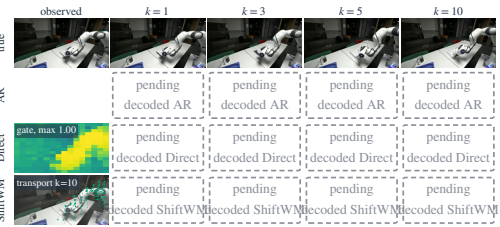


Figure 10: **Transport vs. optical flow.**

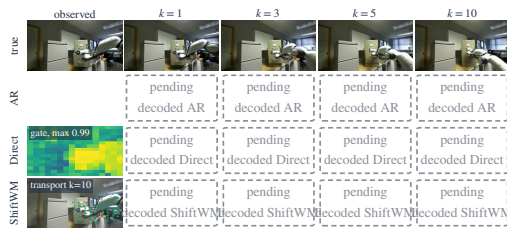
**DROID #1 (0ecbb211b9,  $t = 25$ )**



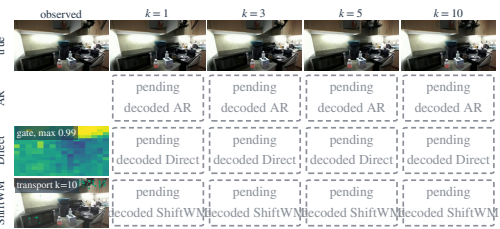
**DROID #2 (7c3480cbfd,  $t = 30$ )**



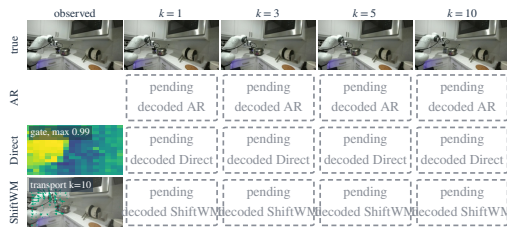
**DROID #3 (d50979cbe4,  $t = 10$ )**



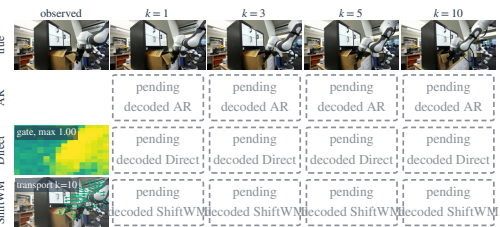
**DROID #4 (bcf4edd11c,  $t = 24$ )**



**DROID #5 (c70170b3ca,  $t = 36$ )**

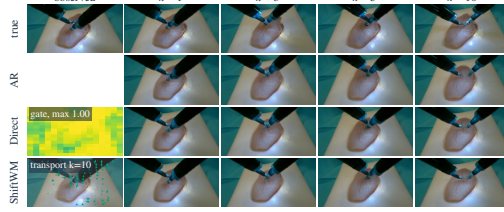


**DROID #6 (b730a44080,  $t = 12$ )**

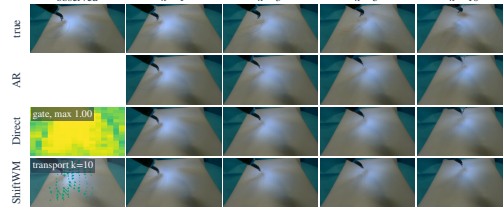


**Figure 11: DROID test rollouts (random selection).**

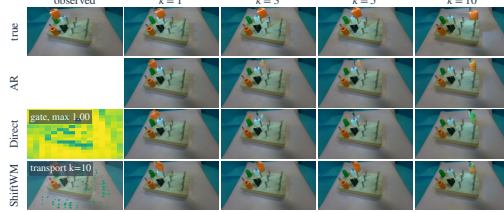
knot tying (ep 000039,  $t = 22$ )



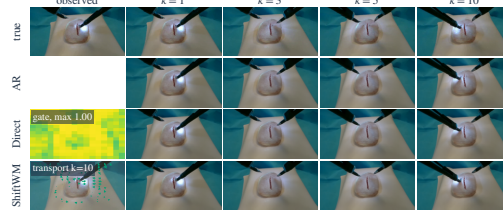
needle grasp and handover (ep 000022,  $t = 31$ )



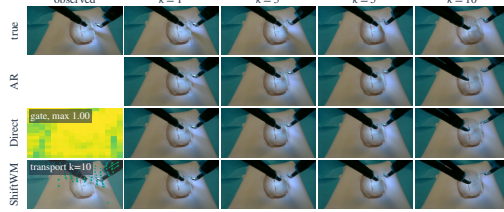
peg transfer (ep 000083,  $t = 21$ )



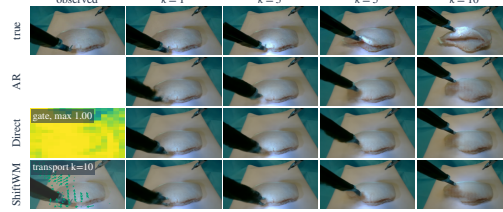
suturing 1 (ep 000042,  $t = 22$ )



suturing 2 (ep 000083,  $t = 72$ )



tissue lifting (ep 000071,  $t = 9$ )



tissue retraction (ep 000053,  $t = 9$ )

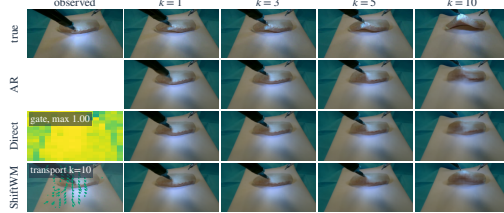


Figure 12: Surgical test rollouts (one per task, random selection).

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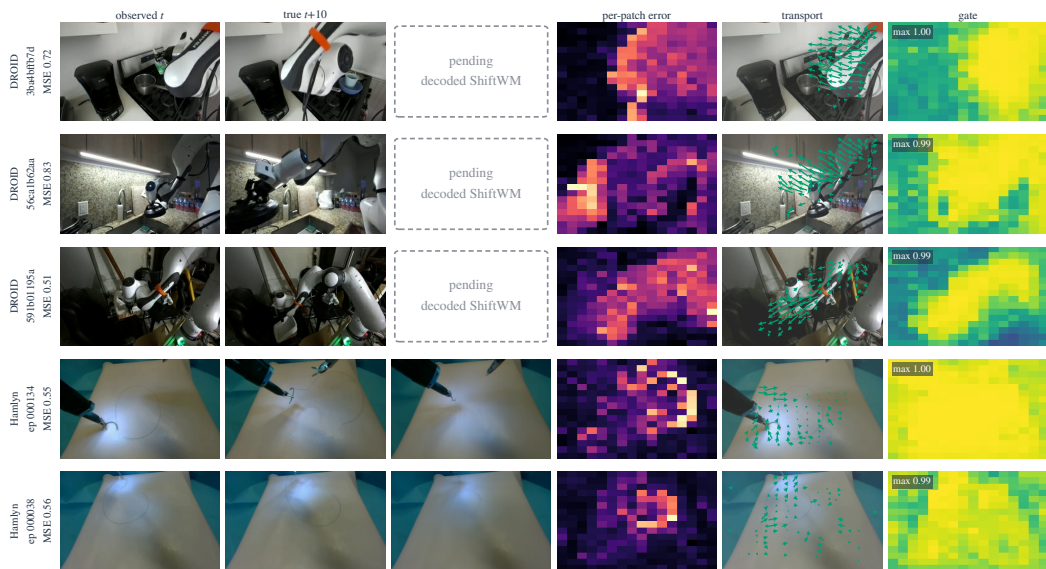


Figure 13: Failure cases.